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| **Semester: III** | **Year: AY 2026-27** |
| **Subject: Programming with Java** | |
| **ASSIGNMENT No : 22** | |
| **TITLE: JDBC PreparedStatement, and ResultSet** | |

### Problem Statement

Demonstration and exploration of JDBC classes and interfaces (PreparedStatement, and ResultSet).

### Learning Objectives

To implement PreparedStatement and ResultSet for secure query execution and data retrieval.

1. Create a login application using PreparedStatement to check username and password.

### Programme

Welcome | databaseConnectionStatus.java | UserLoginApp.java

UserLoginApp.java > UserLoginApp > main(String[])

```java
import java.util.Scanner;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;

public class UserLoginApp {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print(s: "Enter Username (try 'admin'): ");
        String username = scanner.nextLine();
        System.out.print(s: "Enter Password (try 'admin123'): ");
        String password = scanner.nextLine();

        String url = "jdbc:mysql://localhost:3306/rikishab";
        String dbUser = "root";
        String dbPass = "YourNewPassword123!";

        String query = "SELECT * FROM users WHERE username = ? AND password = ?";

        try (Connection conn = DriverManager.getConnection(url, dbUser, dbPass);
             PreparedStatement pstmt = conn.prepareStatement(query)) {

            pstmt.setString(parameterIndex: 1, username);
            pstmt.setString(parameterIndex: 2, password);

            ResultSet rs = pstmt.executeQuery();

            if (rs.next()) {
                System.out.println("Login Successful! Welcome, " + username + ".");
            } else {
                System.out.println(x: "Invalid Username or Password.");
            }

        } catch (Exception e) {
            e.printStackTrace();
        }
```

```
        scanner.close();
    }
}
```

**Output**

```
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>  c:; cd 'c:\Users\RIKISHA\PROGRAM
Temp\cp_eoalbjrpbr5egr0wencuomc3p.argfile' 'UserLoginApp'
Enter Username (try 'admin'): admin
Enter Password (try 'admin123'): admin123
Login Successful! Welcome, admin.
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

2. Create a Java application using PreparedStatement to authenticate hospital staff login (doctor/nurse/login ID and password). Display appropriate access messages based on authentication result.

**Programme**

```
import java.util.Scanner;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;

public class HospitalLoginApp {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter Staff Login ID (try 'HOSP101'): ");
        String loginId = scanner.nextLine();
        System.out.print("Enter Password (try 'med123'): ");
        String password = scanner.nextLine();

        String url = "jdbc:mysql://localhost:3306/rikishab";
        String dbUser = "root";
        String dbPass = "YourNewPassword123!";

        String query = "SELECT role, name FROM hospital_staff WHERE login_id = ? AND password = ?";

        try (Connection conn = DriverManager.getConnection(url, dbUser, dbPass);
             PreparedStatement pstmt = conn.prepareStatement(query)) {

            pstmt.setString(parameterIndex: 1, loginId);
            pstmt.setString(parameterIndex: 2, password);

            ResultSet rs = pstmt.executeQuery();

            if (rs.next()) {
                String role = rs.getString(columnLabel: "role");
                String name = rs.getString(columnLabel: "name");
                System.out.println("Authentication Successful!");
                System.out.println("Welcome " + role + " " + name + ". Access granted to Hospital Management System.");
            } else {
                System.out.println("Access Denied: Invalid Staff ID or Password.");
            }

        } catch (Exception e) {
            e.printStackTrace();
        }
        scanner.close();
    }
}
```

**Output**

```
Problems 11   Output   Debug Console   Terminal   Ports

PS C:\Users\RIKISHA\PROGRAMMING\Java\src>  & 'C:\Program Files\Java\jdk-26.0.1\bin\java.e
italLoginApp'
Enter Staff Login ID (try 'HOSP101'): HOSP101
Enter Password (try 'med123'): med123
Authentication Successful!
Welcome Doctor Dr. Smith. Access granted to Hospital Management System.
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

**Conclusion:**

Thus, the JDBC PreparedStatement and ResultSet were used to execute parameterized SQL queries and retrieve database records for login authentication. This assignment helped in understanding secure query execution, username and password verification, and displaying appropriate access messages based on authentication results.